# Optically Tunable Terahertz Metasurface Absorber Based on Fe-Doped β-$Ga_2O_3$

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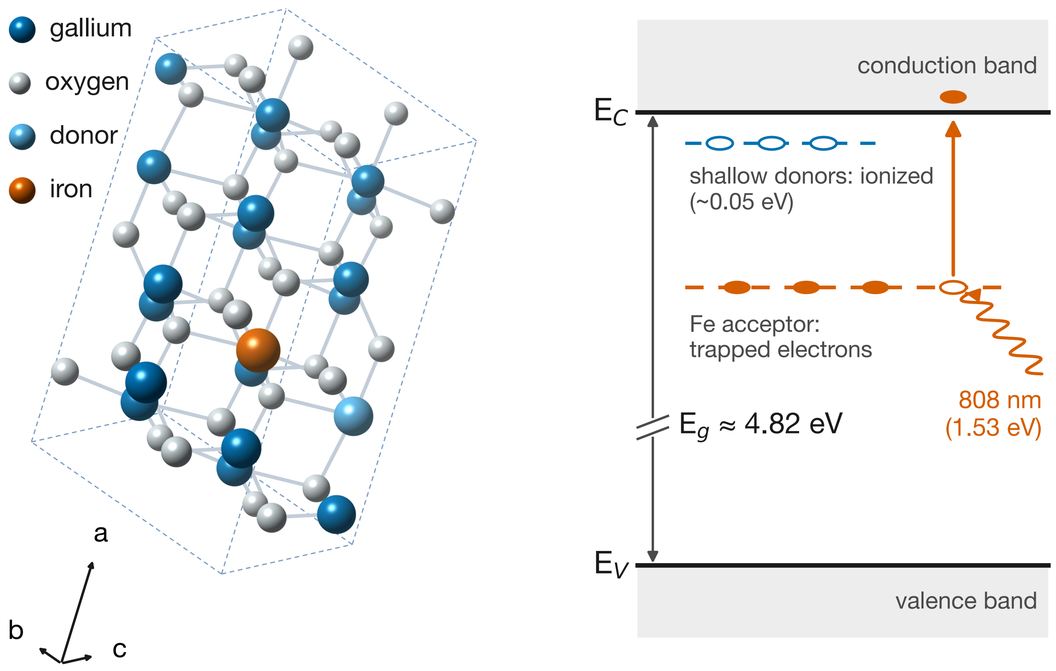
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## Motivation

- Reconfigurable THz surfaces usually need contacts, gates or heaters on every element.
- **Here light tunes the substrate itself:** an 808 nm pump and deep Fe levels in β-$Ga_2O_3$.
- No contacts, and any resonator on top becomes tunable.

## Mechanism




- Fe traps the electrons; 808 nm light frees them: **~25 S/m** [1].

## Measured Photoconductivity

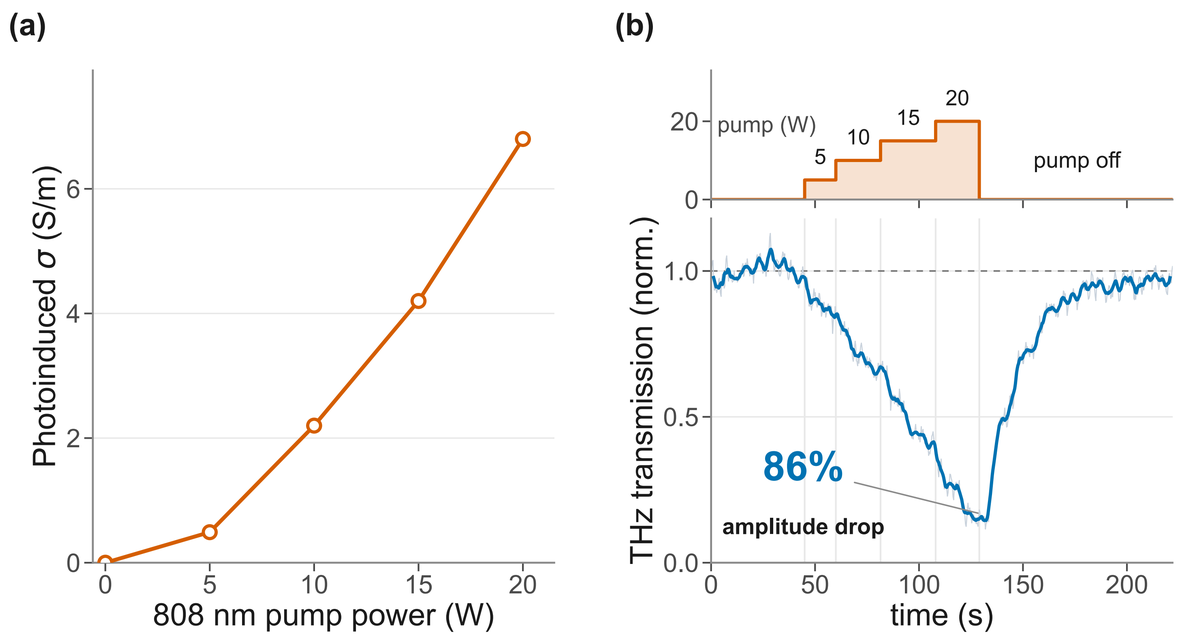



**Figure 1. Measured**, bare crystal. (a) σ vs pump power. (b) 463 GHz, pump to 20 W, then off.

- **86% THz switching**, no metasurface yet.

## H-Resonator Metasurface

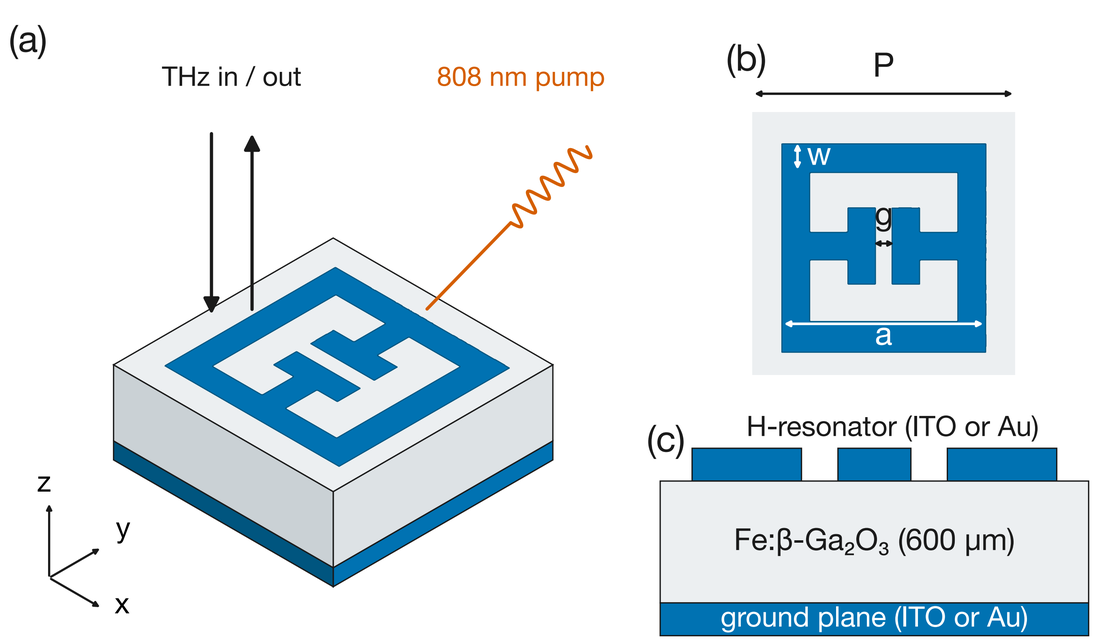



**Figure 2.** Unit cell: P 150, a 120, w 15, g 5 μm [2].

- Ground plane: THz is **reflected or absorbed**, the pump decides. **Gold first, then ITO.**

## Gold / Fe:β-$Ga_2O_3$ Absorber

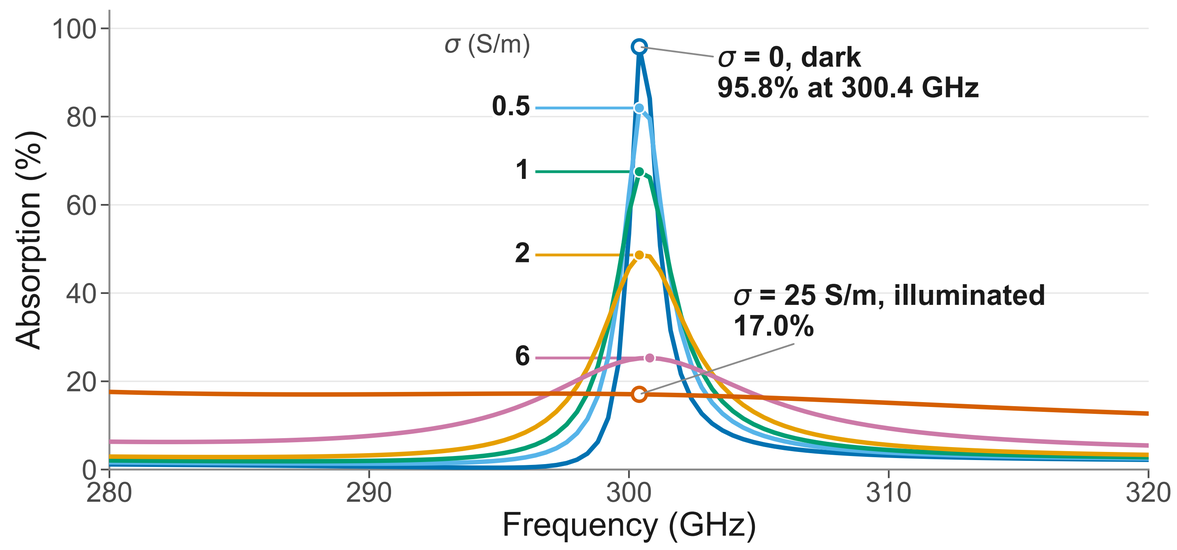



**Figure 3. Simulated.** Au H-resonators on Au ground plane.

| **82.2%** | **95.0%** |
|---|---|
| absorption modulation | reflected-power modulation |

## TE / TM Switching in ITO

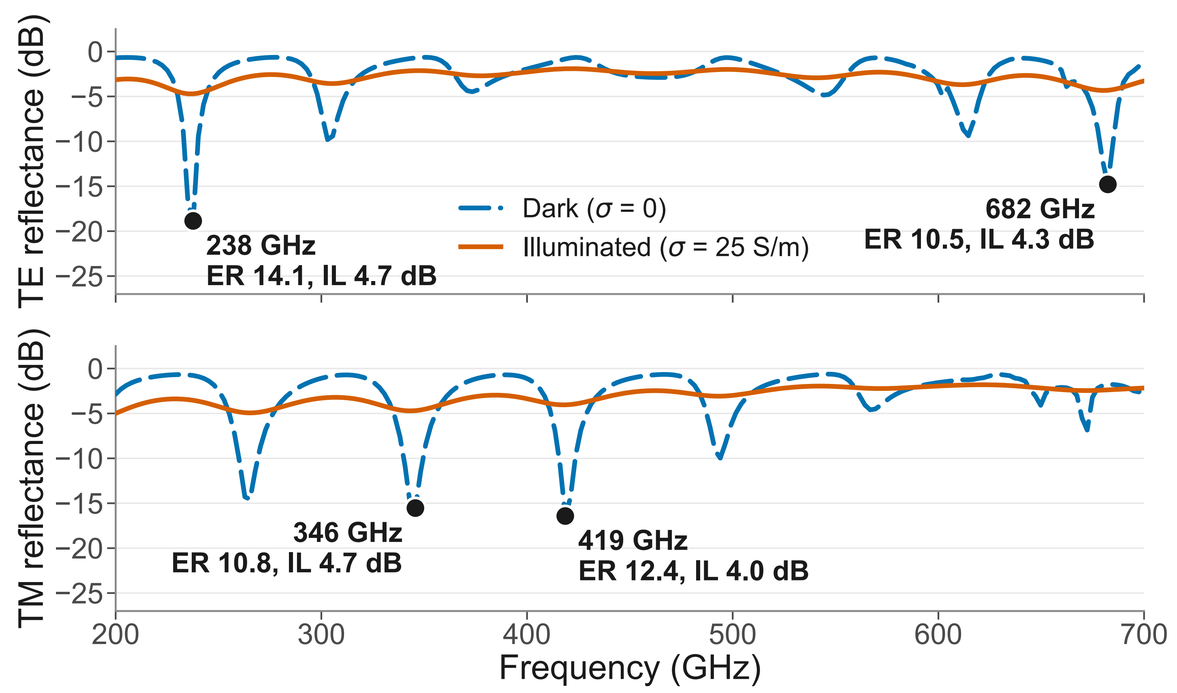



**Figure 4. Simulated.** ITO on (010) Fe:β-$Ga_2O_3$. ER: contrast, IL: loss.

- **Four channels ≥ 10 dB**, two per polarization.

## Fabricated Device

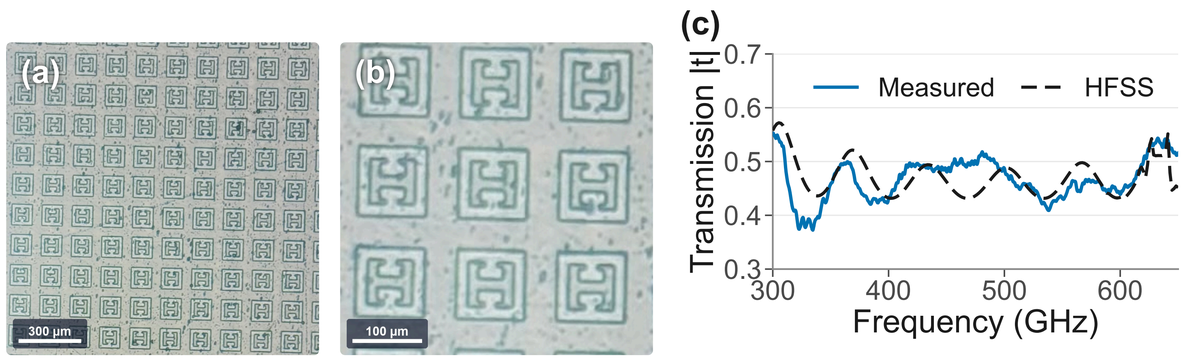



**Figure 5.** (a) ITO H-resonator array; (b) a 99 / w 15.5 / g 19 μm, 20 Ω/sq. (c) **Measured** transmission vs HFSS: rms 0.04.

- **Preliminary, in transmission:** no anneal, no ground plane; already matches the simulation.
- Every cell survived lift-off. Next: anneal, ground plane, pump.

*LOR 10B / 1813 · laser writer · 2.5 μm undercut · sputtered ITO · lift-off*

## Conclusions

- **Measured:** 86% switching, no contacts.
- **Simulated:** 82.2% absorption modulation (gold); four ITO channels ≥ 10 dB.
- **Fabricated:** ITO array matches HFSS in transmission.

### References

[1] H. Jiang *et al.*, *Photonics* **9**, 233 (2022).
[2] R. Degl'Innocenti *et al.*, *ACS Nano* **8**, 2548 (2014).

### Acknowledgment

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